

Ingredient Detection and Recipe Recommendation System Using Artificial Intelligence

Project Report

V. Nikhil, Chavva Sai Nihal, K.Rushik

Amrita Vishwa Vidyapeetham

Department of Electronics and Communication Engineering

Abstract

We developed an Ingredient Detection and Recipe Recommendation System that uses deep learning and computer vision to automatically identify food items stored in a refrigerator and suggest relevant recipes. The system is built around YOLOv8, fine-tuned on a custom dataset of refrigerator images collected via Roboflow. The trained model can detect a variety of food items including vegetables, fruits, eggs, dairy products, and beverages — from a single image, achieving precision ≈ 0.95 , recall ≈ 0.98 , $mAP@50 \approx 1.00$, and $mAP@50-95 \approx 0.85$ after 50 training epochs. The detected ingredients are then matched against a recipe database to suggest dishes that can be prepared with what is available, helping reduce food wastage and making everyday cooking decisions simpler.

Keywords — YOLOv8, object detection, ingredient detection, recipe recommendation, deep learning, Roboflow, food management

I. INTRODUCTION

A common problem in most households is figuring out what to cook based on what is actually available in the fridge. It is easy to lose track of stored food items, which often leads to buying groceries that are already at home, ingredients going bad before they are used, and unnecessary wastage. Looking up recipes manually or searching online without knowing what you have is both inefficient and frustrating.

Recent progress in Artificial Intelligence and computer vision has made it possible to tackle such practical, everyday challenges more effectively. Modern object detection models can recognize and locate multiple items within an image in real time — a capability that maps naturally onto the problem of identifying ingredients inside a refrigerator.

In this work, we built an Ingredient Detection and Recipe Recommendation System using YOLOv8, fine-tuned on a domain-specific Roboflow dataset of fridge images. Our main contributions are: (1) creating and training YOLOv8 on a fridgespecific ingredient dataset; (2) achieving $mAP@50$ close to 1.00; (3) connecting the detection output to a recipe recommendation module; and (4) validating the system on real fridge images to confirm practical usability.

II. METHODOLOGY

Our pipeline consists of four stages: dataset preparation, model training, object detection, and recipe recommendation.

A. Dataset Preparation

We sourced our training dataset from Roboflow, which is a widely used platform for managing and augmenting computer vision datasets. Images of refrigerator interiors were collected and manually annotated with bounding boxes. The dataset covers eight ingredient categories: oranges, eggs, lettuce, cucumbers, tomatoes, milk, orange juice, and parsley. To make the model more robust against real-world variations, we applied standard augmentation

techniques including horizontal flipping, brightness and contrast adjustments, and mosaic augmentation.

B. Model Architecture – YOLOv8

YOLOv8, developed by Ultralytics, is a single-stage object detector that processes an entire image in one forward pass to predict bounding boxes and class probabilities. Unlike older YOLO versions, YOLOv8 uses an anchor-free detection head and a decoupled architecture that handles classification and bounding box regression independently. This design leads to faster convergence and better accuracy. We started with a COCO-pretrained YOLOv8 model and fine-tuned it on our custom fridge dataset.

Training is supervised using three loss components:

- Box Loss: measures how well the predicted bounding box coordinates match the ground truth.
- Classification Loss: penalizes the model when it assigns the wrong class label to a detected object.
- DFL Loss: Distribution Focal Loss, which helps improve the precision of bounding box boundary predictions.

C. Training Configuration

We trained the model for 50 epochs. The key configuration details are listed in Table I.

Parameter	Value
Model	YOLOv8
Base Weights	COCO Pretrained
Epochs	50
Dataset Source	Roboflow
Optimizer	SGD / AdamW

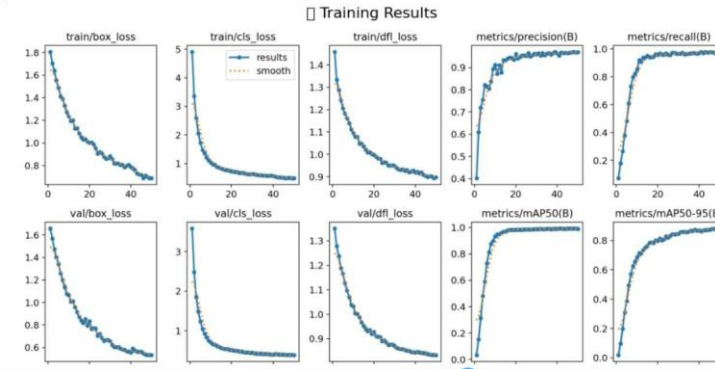
Augmentation	Mosaic, Flip, Brightness
Eval Metric	mAP@50, mAP@50-95

TABLE I. Training Configuration Parameters

D. Recipe Recommendation Module

Once YOLOv8 finishes detecting ingredients in the fridge

Fig. 1 shows the training progress over all 50 epochs. All three loss curves — box loss, classification loss, and DFL loss — decrease steadily and consistently, which indicates that training was stable and the model was learning without overfitting. The validation losses closely mirror the training losses throughout, confirming that the model generalizes well to unseen data.



image, the identified items are assembled into a list. This list is then compared against a pre-built recipe database to find dishes that can be made with those ingredients. Recipes are ranked based on how many of their required ingredients are already available, with priority given to recipes that need the least number of additional items to be purchased.

III. EXPERIMENTAL RESULTS

The model was trained for 50 epochs, and both loss curves and performance metrics were tracked at every epoch to monitor progress.

A. Training and Validation Loss

Fig. 1. YOLOv8 Training Results – Loss curves and performance metrics over 50 epochs.

B. Performance Metrics

Table II shows the final performance metrics recorded at the end of epoch 50.

Metric	Final Value
Precision (B)	~0.95
Recall (B)	~0.98
mAP@50 (B)	~1.00
mAP@50-95 (B)	~0.85
Train Box Loss	~0.68

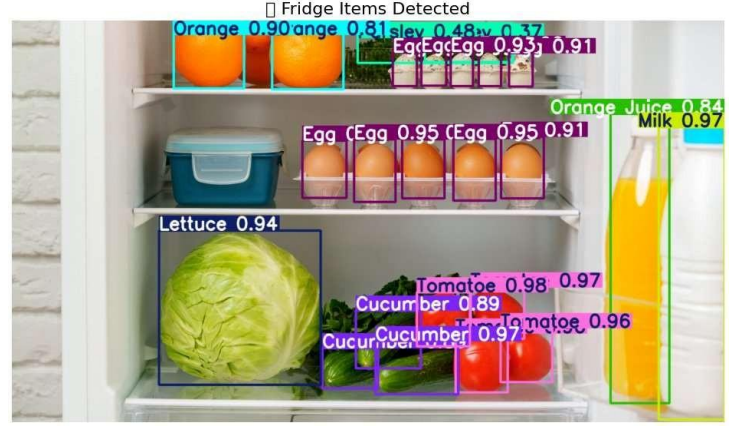


Fig. 2. YOLOv8 detection on a test fridge image with bounding boxes and confidence scores.

Train Cls Loss	~0.60
Train DFL Loss	~0.88
Val Box Loss	~0.56
Val Cls Loss	~0.55
Val DFL Loss	~0.86

TABLE II. Final Model Performance Metrics at Epoch 50

A precision of ~0.95 and recall of ~0.98 mean the model is both accurate and thorough — it correctly identifies nearly every ingredient present while keeping false positives low. The mAP@50 of nearly 1.00 shows that bounding box predictions are well-aligned with the actual ingredient locations. An mAP@50-95 of ~0.85 further confirms that the model maintains good accuracy even when evaluated at stricter IoU thresholds, making it suitable for real-world use.

C. Detection on Test Image

Fig. 2 shows the detection output on a real refrigerator image that was not part of the training set. The model correctly identifies multiple items at once: Oranges (0.90, 0.81), Parsley (0.48), Eggs (0.93, 0.95, 0.91), Orange Juice (0.84), Milk (0.97), Lettuce (0.94), Cucumbers (0.89, 0.97), and Tomatoes (0.98, 0.97, 0.96). Most items are detected with high confidence, demonstrating that the system works reliably on images it has never seen before.

IV. DISCUSSION

Our results confirm that YOLOv8, trained on fridgespecific data, is well-suited for multi-object ingredient detection in a home setting. The smooth drop in both training and validation losses shows that the model was learning consistently throughout training, and the close alignment between the two curves rules out overfitting a direct benefit of the data augmentation we applied.

The near-perfect mAP@50 score tells us the model is reliably placing bounding boxes in the right locations. Even at stricter IoU thresholds (mAP@50-95 of ~0.85), performance remains strong, which gives us confidence that the system would hold up well in real deployment scenarios.

Common items like Milk (0.97), Tomatoes (0.98), and Eggs (0.95) were detected with particularly high confidence, which reflects how well-represented they are in the training data. Parsley scored lower at 0.48, likely because fewer annotated examples were available for it — adding more training images for such classes should improve their detection in future iterations.

V. FUTURE SCOPE

There are several directions in which this system can be extended:

1. Reading expiry dates from labels using OCR.
2. Estimating the quantity or volume of each detected ingredient.
3. Automatically generating a shopping list for any ingredients a recipe needs but are not currently in the fridge.
4. Deploying the system as a mobile app to enable on device, real-time detection.
5. Adding voice assistant support for a hands-free cooking experience.
6. Showing nutritional information alongside the recommended recipes.
7. Cloud integration for storing fridge history and maintaining personalized dietary profiles per user.

Looking ahead, exploring transformer-based detectors like RT-DETR or integrating multi-modal large language models could further push detection accuracy and make the recipe recommendation module significantly more intelligent.

VI. CONCLUSION

In this project, we built an Ingredient Detection and Recipe Recommendation System using YOLOv8 trained on a Roboflow dataset of real refrigerator images. The final model achieved precision ~0.95, recall ~0.98, mAP@50 ~1.00, and mAP@5095 ~0.85 after 50 epochs of training. Testing on real fridge

photographs confirmed that the model can detect several ingredients simultaneously, with high confidence scores across most categories.

By linking the detection output to a recipe recommendation module, the system provides a genuinely useful solution to a common household problem — cutting down on manual effort and helping reduce food wastage. We believe this project demonstrates how deep learning and computer vision can be applied to create practical, everyday tools that make a real difference.

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